

Chapter 1

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Problem 8.1.

Show that an inverse limit of nonempty finite discrete sets (for any directed index set) is nonempty. [*Hint:* For such an inverse system $\{X_\alpha : \alpha \in \Lambda\}$, consider the subsets $X_{\lambda\mu} \subseteq \prod X_\alpha$ consisting of the sequences (x_α) satisfying $\phi_{\lambda\mu}(x_\mu) = x_\lambda$ for a fixed pair $\lambda \leq \mu$ and use the compactness of the topological product of the X_α . We shall go through this argument in a more general case in Lemma 3.4.12.]

Proof.

Let $\{X_\alpha : \alpha \in \Lambda\}$ denote such an inverse system, and let $X_{\lambda\mu} \subseteq \prod X_\alpha$ be the set of sequences (x_α) with $\phi_{\lambda\mu}(x_\mu) = x_\lambda$ for some $\lambda \leq \mu$. This is a closed set, as its complement admits the open cover $\prod_{\alpha \neq \lambda, \mu} X_\alpha \times \{x_\mu\} \times (X_\lambda \setminus \{x_\lambda\})$. We deduce that $X_{\lambda\mu}$ is itself a compact space as a closed subset of a compact space.

On the other hand, by the compatibility condition of inverse systems, the collection of all such $X_{\lambda\mu}$'s must then satisfy the finite intersection property. Hence the intersection of all the $X_{\lambda\mu}$'s must be nonempty. But then this is exactly the inverse limit of the system. ■

Problem 8.2.

Let G be a profinite group, p a prime number. A **pro- p -Sylow subgroup** of G is a pro- p -group whose image in each finite quotient of G is of index prime to p . Show that pro- p -Sylow subgroups exist and they are conjugate in G . [*Hint:* Apply the previous exercise to the inverse system formed by the sets of p -Sylow subgroups in each finite quotient of G .]

Proof.

Consider the directed system of finite quotients of G . Every finite quotient has finitely many Sylow p -subgroups, the collection of which is a finite set that we can endow with the discrete topology. By the correspondence theorem, pullbacks of Sylow p -subgroups satisfy the cocycle condition, such that this gives an induced directed system of sets of Sylow p -subgroups in each finite quotient of G . But then by the result of the previous question, the inverse limit of this set is nonempty. But then every Sylow p -subgroup of a finite quotient of G corresponds to a subgroup of G , such that the inverse limit of the system of Sylow p -subgroups corresponds to an inverse limit of a system of subgroups of G . Hence the result is indeed a well-defined pro- p -Sylow subgroup of G .

Now, suppose that we have two such pro- p -Sylow subgroups P_1, P_2 . They are conjugate in each finite quotient of G , such that the quotient of a conjugating element is still a conjugating element. Hence we obtain another inverse system of conjugating elements in G , which is nonempty and corresponds to an element of G that carries P_1 to P_2 by the same reasoning. ■

Problem 8.3.

Let k be a perfect field, p a prime number. Show that there exists an algebraic extension $k^{(p)} \mid k$ such that each finite subextension is of degree prime to p and $k^{(p)}$ has no nontrivial finite extensions of degree prime to p . Is such an extension unique inside a fixed algebraic closure? [Hint: Use the previous exercise.]

Proof.

As k is a perfect field, $\bar{k} = k^{\text{sep}}$. Denote the absolute Galois group by G . If K is some finite degree p -coprime extension of k , let L be the normal closure of K in $\bar{k}$ such that we get a finite quotient $G \twoheadrightarrow \text{Gal}(L/k)$. Then K corresponds to a Sylow p -subgroup of $\text{Gal}(L/k)$. In other words, the inverse system of finite p -coprime extensions of k corresponds to the inverse system of Sylow p -subgroups in finite quotients of G . Hence by the previous result this induces a well-defined pro- p -Sylow subgroup that corresponds to a maximal p -coprime extension $k^{(p)}$ of k by the fundamental theorem of Galois theory. In particular, as pro- p -Sylow subgroups are only unique up to conjugation, $k^{(p)}$ will only be unique up to Galois conjugation. ■

Problem 8.4.

Consider the compositum E of all quadratic extensions of $\mathbb{Q}$ inside a fixed algebraic closure $\bar{\mathbb{Q}}$.

(a) Show that $\text{Gal}(E|\mathbb{Q})$ is uncountable and has uncountably many subgroups of index 2.

Proof.

Let $T = \{-1\} \cup \{\text{primes}\}$, such that then E is generated over $\mathbb{Q}$ by $\{\sqrt{t} : t \in T\}$, and the extensions $\{\mathbb{Q}(\sqrt{t})\}$ are linearly disjoint over $\mathbb{Q}$. Every $\sigma \in \text{Gal}(E/\mathbb{Q})$ is therefore uniquely determined by the signs $\sigma(\sqrt{t}) = \pm\sqrt{t}$ for every $t \in T$. Hence $\text{Gal}(E/\mathbb{Q}) \simeq \prod_{t \in T} \mathbb{Z}/2\mathbb{Z} \simeq (\mathbb{Z}/2\mathbb{Z})^{\aleph_0}$, which has cardinality $2^{\aleph_0}$ and is uncountable. Moreover, as the infinite product of abelian groups is abelian, it immediately follows that the group admits uncountably many quotients into $\mathbb{Z}/2\mathbb{Z}$; i.e. has uncountably many subgroups of index 2. ■

(b) Deduce that there are uncountably many subgroups of index 2 in $\text{Gal}(\bar{\mathbb{Q}}|\mathbb{Q})$ that are not open.

Proof.

All degree 2 extensions of $\mathbb{Q}$ are necessarily primitive and Galois. Hence we can write $E = \mathbb{Q}(\alpha)$ for some α , such that the extension is entirely determined by the minimal polynomial of α . Hence we get a surjection from the set of degree 2 irreducible polynomials in $\mathbb{Q}[x]$ to the set of quadratic extensions of $\mathbb{Q}$. So both sets must be countable. By the pigeonhole principle, and by the result of (a), this implies that there are uncountably many subgroups of index 2 in $\text{Gal}(\bar{\mathbb{Q}}|\mathbb{Q})$ that do not correspond to intermediate fields; i.e. are not open (as a closed finite index subgroup is open - Lemma 1.3.8.). ■

Problem 8.5.

Let G be a profinite group acting via field automorphisms on a field K . Assume that the action is continuous when K carries the discrete topology and that each nontrivial element in G acts nontrivially on K . Show that $G \cong \text{Gal}(K|k)$, where $k = K^G$.

Proof.

As G acts via k -fixed field automorphisms of K , there is a well-defined map $G \hookrightarrow \text{Aut}(K/k)$. The fact that each nontrivial element in G acts nontrivially on K guarantees that this map is an embedding. Moreover note that $\text{Fix}(\text{Aut}(K/k)) \subseteq \text{Fix}(G) = k$, so by Definition 1.2.1. (L/k is Galois iff $\text{Fix}(\text{Aut}(L/k)) = k$) K/k is indeed Galois.

Next, as K is assumed to carry the discrete topology, the action being continuous is equivalent to the stabilizers G_x of every point $x \in K$ being open in G (see p.20). This implies that the embedding $G \hookrightarrow \text{Gal}(K/k)$ is in fact a continuous map: Open sets in $\varprojlim \text{Gal}(L/k) \cong \text{Gal}(K/k)$ are of the form $U = \{\sigma : \sigma|_L \subseteq U_L\}$, where $U_L \subseteq \text{Gal}(L/k)$ is some arbitrary subset and L is finite degree. By the primitive element theorem, $L = k(\alpha)$ for some α . Then elements of U_L correspond bijectively to elements of the orbits of α ; such that U is equal to the finite union of the stabilizers of those elements; and its preimage in G is open.

Finally notice that G and $\text{Gal}(K/k)$ are both compact Hausdorff groups; such that the image of G in $\text{Gal}(K/k)$ is necessarily closed. To complete the proof; notice that by the fundamental theorem of Galois theory we get that G is the unique closed subgroup of $\text{Gal}(K/k)$ that fixes k ; i.e. $\text{Gal}(K/k)$ itself. So the map is a group isomorphism. The map is also a homeomorphism, as it is a bijective continuous map between compact Hausdorff spaces. ■

Problem 8.6.

(Leptin, Waterhouse) Show that every profinite group G arises as the Galois group of some Galois extension $K|k$. [Hint: For each open normal subgroup $N \triangleleft G$ fix a system of left coset representatives $1 = \sigma_1^N, \dots, \sigma_m^N$. Let F be a perfect field, and $K|F$ the purely transcendental extension obtained by adjoining an indeterminate x_i^N for each σ_i^N . Make G act on F trivially, and on K via $\sigma(x_i^N) = x_j^N$, where x_j^N corresponds to σ_j^N with $\sigma_j^N N = \sigma(\sigma_i^N N)$. Verify that this action satisfies the criterion of the previous exercise.]

Remark. The statement does not hold if one requires K to be a separable closure of k . For instance, Artin and Schreier showed in [3] that among the nontrivial finite groups only $\mathbb{Z}/2\mathbb{Z}$ can arise as an absolute Galois group.

Proof.

That the action is well-defined follows from construction. We show via an application of Problem 8.5 that G is isomorphic to $\text{Gal}(K/K^G)$ as topological groups.

The action is continuous: By the same reasoning as in Problem 8.5, this is equivalent to showing that G_x is open for all $x \in K$. x is then necessarily a rational function in finitely many of the x_i^N 's, so there exists finitely many open normal subgroups $N_1, \dots, N_n$ such that $x \in F(x_1^{N_1}, \dots, x_m^{N_1}, \dots, x_1^{N_n}, \dots, x_m^{N_n})$. Then G_x contains the open (normal) subgroup $\bigcap N_i$; such that G_x is itself necessarily normal as it is a union of the cosets of $\bigcap N_i$.

The action is faithful: This amounts to showing that the only element in G that is trivial in G/N for each open normal subgroup N is the identity. To that end we show that such an element g is necessarily contained in every neighbourhood of the identity; and as the topology on profinite groups is Hausdorff we will be done. By (p.12) we have that the kernels of the projections $G \rightarrow G_i$ form an open neighbourhood basis of the identity. But then g necessarily is in the same neighbourhood of 1 for every element of this basis. ■

Problem 8.7.

Let k be a field, and A a finite étale k -algebra equipped with an action of a finite group G via k -algebra automorphisms; we call such algebras G -algebras. We moreover say that A is *Galois* with group G if $\dim_k(A)$ equals the order of G and $A^G = k$.

(a) Consider the G -algebra structure on $A \otimes_k \bar{k}$ given by $g(a \otimes \alpha) = g(a) \otimes \alpha$. Prove that A is Galois with group G if and only if $A \otimes_k \bar{k}$ is isomorphic to the group algebra $\bar{k}[G]$ as a G -algebra.

Proof.

For the following proof, we write $\bar{k}^{|G|}$ instead of $\bar{k}[G]$ to avoid notational confusion.

($\Leftarrow$) : Suppose that $A \otimes_k \bar{k} \cong \bar{k}^{|G|}$. As the dimension of a vector space remains unchanged under extension of scalars, we have that $\dim_k A = \dim_{\bar{k}} A \otimes_k \bar{k} = \dim_{\bar{k}} \bar{k}^{|G|} = |G|$ (this argument only needs the isomorphism as k -algebras).

Now, as the action of G on $A \otimes_k \bar{k}$ is defined to be $g(a \otimes \alpha) = g(a) \otimes \alpha$, we have that $A^G \otimes_k \bar{k} = (A \otimes_k \bar{k})^G = (\bar{k}^{|G|})^G = \bar{k}$ (this argument requires the isomorphism to be G -equivariant as well!). By the same reasoning as above we have that $\dim_k A^G = \dim_{\bar{k}} A^G \otimes_k \bar{k} = 1$; so $A^G = k$. A is indeed G -Galois.

($\Rightarrow$) : Suppose that A is G -Galois. Notice that the k -linear G -action on A induces an $\bar{k}$ -linear G -action on $A \otimes_k \bar{k}$; as $g \cdot (a \otimes \alpha) = g(a) \otimes \alpha = \alpha(g \cdot a \otimes 1)$. But then $A \otimes_k \bar{k}$ is isomorphic to $\bar{k}^{\dim_k A} = \bar{k}^{|G|}$ as G -algebras; where the G -action on $\bar{k}^{|G|}$ is induced via the map. We are hence reduced to finding the $\bar{k}$ -algebra automorphisms of $\bar{k}^{|G|}$.

We show that these are exactly given by the permutations of the coordinates: Let ϕ be such an automorphism, and let I denote an index set for the basis vectors. Then for all $i \in I$, we have $\phi(e_i)^2 = \phi(e_i)$ such that $\phi(e_i) = \sum_{j \in J} e_j$ for some $J \subset I$. But then the $\phi(e_i)$'s must also be pairwise orthogonal so they must have disjoint support; hence $\phi(e_i) = e_j$ for some j .

We now show that G acts (simply) transitively on I . Note that $|G| = |I|$ so simple transitivity and transitivity are equivalent. To that end, notice that $A^G = k$ implies $(A \otimes_k \bar{k})^G = \bar{k}$, which is one dimensional; such that the only nontrivial G -invariant subspace is exactly given by the span of the vector $\sum e_i$. But then G can only have one orbit on I , as there would otherwise be another nontrivial invariant subspace. Hence I is isomorphic to G as left G -sets; and $A \otimes_k \bar{k}$ is isomorphic to $\bar{k}^{|G|}$ as G -algebras. ■

(b) Making G act on $\text{Hom}_k(A, k_s)$ via $\phi \mapsto \phi \circ g$, show that in the correspondence of Theorem 1.5.4, Galois algebras with group G correspond to finite continuous $\text{Gal}(k)$ -sets with simply transitive G -action.

Proof.

For the following proof, we write $\bar{k}^{|G|}$ instead of $\bar{k}[G]$ to avoid notational confusion.

As A is already an étale algebra, the statement of Theorem 1.5.4. already gives us that $\text{Hom}_k(A, k_s)$ is a finite continuous $\text{Gal}(k)$ -set. Hence we need only verify that the induced G -action is transitive.

Notice that $\bar{k}$ -algebra homomorphisms $A \otimes_k \bar{k} \cong \bar{k}^{|G|} \rightarrow \bar{k}$ are given exactly by coordinate projection maps: The idempotents e_i must be sent to 0 or 1 only; but the fact that $1 = \sum e_i$ is sent to 1 enforces that exactly one of the e_i 's is sent to 1. Hence G acts transitively on the set $\text{Hom}_{\bar{k}\text{-alg}}(A \otimes_k \bar{k}, \bar{k})$.

This implies that G acts transitively on $\text{Hom}_k(A, k_s)$: as $k \subset \bar{k}$ is a field extension, extension of scalars induces a bijection on hom-sets; such that $\text{Hom}_{k(A, k_s)} = \text{Hom}_k(A, \bar{k}) \cong \text{Hom}_{\bar{k}}(A \otimes_k \bar{k}, \bar{k})$.

Note that the converse is true as well: if A is étale such that $\text{Hom}_k(A, k_s)$ admits a transitive G -action then so does $\text{Hom}_{\bar{k}}(A \otimes_k \bar{k}, \bar{k})$. But then $A \otimes_k \bar{k}$ is isomorphic to $\bar{k}^{\dim_k A}$ as $\bar{k}$ -algebras, such that by a similar argument in (a) one can show that the simple transitivity enforces that $A \otimes_k \bar{k} \cong \bar{k}^{|G|}$ such that A is G -Galois. ■

Problem 8.8.

Let k be a field of characteristic different from 2, and S a continuous left $\text{Gal}(k)$ -set with n elements. Consider the subset $\Sigma(S) \subset S^n$ consisting of n -tuples $(s_1, \dots, s_n)$ with $s_i \neq s_j$ for $i \neq j$. It inherits a continuous left action of $\text{Gal}(k)$ from the product action on S^n , and it also has a natural action by the symmetric group S_n via permutation of the components. Denote by $\Delta(S)$ the quotient of $\Sigma(S)$ by the action of the alternating group $A_n \subset S_n$. It is a 2-element continuous left $\text{Gal}(k)$ -set.

(a) Show that the finite étale k -algebra corresponding to $\Delta(S)$ via Theorem 1.5.4 is isomorphic to $k \times k$ if $\text{Gal}(k)$ acts on $\Delta(S)$ by even permutations, and is a degree 2 field extension of k otherwise.

Proof.

Suppose that $\text{Gal}(k)$ acts by even permutations. Then the action has exactly two orbits, so the corresponding étale algebra is just a direct product of the stabilizers subfields of the two elements. But then notice that as each orbit is a one element set, the stabilizer is necessarily the entirety of $\text{Gal}(k)$. Under the fundamental theorem this corresponds to k ; i.e. the action corresponds to $k \oplus k$.

Now suppose that the action of $\text{Gal}(k)$ has at least one odd permutations. Then the action has one orbit on the two-element set, so this $\text{Gal}(k)$ -set is isomorphic to $\text{Hom}(L, k_s)$ for some finite separable extension L . This set must have size 2, such that L itself must be a degree 2 field extension of k . ■

(b) If A is the finite étale k -algebra corresponding to S via Theorem 1.5.4, denote the k -algebra of (a) by $\Delta(A)$. Show that when $A \cong k[x]/(f)$ for a polynomial f without multiple roots, then $\Delta(A) \cong k[x]/(x^2 - d(f))$, where $d(f) \in k$ is the Vandermonde determinant formed from the roots of f .

Remark. The k -algebra $\Delta(A)$ is called the *discriminant* of the finite étale k -algebra A . For a description of $\Delta(A)$ in the general case, see [45], Proposition 18.24.

Proof.

For the following proof, we take the definition of the Vandermonde determinant to be $d(f) := \prod_{i < j} (\alpha_i - \alpha_j)$. I suspect the author of the question is actually referring to the discriminant $d(f)^2$, so keep track of the discrepancy in notation.

When $A \cong k[x]/(f)$ for some separable f , A is a $\deg f$ -dimensional product of separable extension of k ; and the action of $\text{Gal}(k)$ on S is isomorphic to the action of $\text{Gal}(k)$ on the roots of f . But then the Vandermonde determinant is an alternating form; such that $\sigma \cdot d(f) = \text{sgn}(\sigma)d(f)$ for all permutations σ . Hence $\Delta(S)$ is isomorphic as $\text{Gal}(k)$ -sets to the set $\{d(f), -d(f)\}$, which is indeed a 2-element set as the characteristic is not 2. But then the action of $\text{Gal}(k)$ is odd $\iff$ the action is transitive $\iff$ the $\text{Gal}(k)$ -conjugates of $d(f)$ are $\pm d(f)$ $\iff d(f)$ is not in k . So $k[x]/(x^2 - d(f)^2)$ is a quadratic extension of k iff $\text{Gal}(k)$ is odd, and is isomorphic to $k[x]/(x + d(f))(x - d(f)) \cong k \oplus k$ otherwise. By (a) we are done. ■